

Kelly Buabeng

Backend Engineer | Data Science & ML | Cybersecurity

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PROFESSIONAL SUMMARY

Results-driven Backend Software Engineer and aspiring Data Scientist in the final year of a BSc Mathematical Sciences (Computer Science with Statistics) degree at the University of Ghana. Proven experience building and optimising backend systems, implementing RESTful APIs, and supporting enterprise IT infrastructure across high-profile organisations including Xavs Labs, TotalEnergies Marketing Ghana PLC, and the Ghana Civil Aviation Authority. Recognised as a Top 2 Finalist (Ghana) in the 2025 NASA International Space Apps Challenge for delivering a machine learning solution on NASA's Kepler exoplanet dataset. Currently pursuing an Applied Data Science Certificate from WorldQuant University to deepen expertise in statistical modelling, machine learning, and data analytics. Committed to building secure, scalable, and data-driven solutions for African and global markets.

CORE COMPETENCIES

Backend Dev: REST APIs, Node.js, Python Flask	ML & Computer Vision: Scikit-learn, OpenCV, Pandas, NumPy	Cybersecurity: Threat Analysis, Security Protocols, Vulnerability Assessment
Data Science: Statistical Modelling, Feature Engineering, Jupyter	Databases: SQL optimisation, Relational DB Design	DevOps & Deploy: Docker, Git & GitHub, Vercel, CLI Tools
Systems: Linux, System Administration	Frontend: React, Next.js	Soft Skills: Technical Documentation, Agile Collaboration

TECHNICAL STACK

Languages:	Python, JavaScript (Node.js), Prolog
Frameworks:	Flask, React, Next.js
ML & CV:	Scikit-learn, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebooks
Python Libs:	Requests, SQLAlchemy, Pydantic, dotenv, Pytest, FastAPI
APIs & Tools:	REST APIs, Git & GitHub, CLI Tools, Vercel, Postman
DevOps:	Docker (containerisation & deployment), Linux
Databases:	SQL (query optimisation), relational database design
Cybersecurity:	Vulnerability Assessment, Risk Analysis, Security Protocol Implementation, Cryptography

WORK EXPERIENCE

Backend Developer Intern | Xavs Labs

Aug 2025 – Jan 2026

Accra, Ghana

- ▶ Built and maintained backend services for web applications using Python and modern frameworks, contributing to production-grade code bases.
- ▶ Designed and implemented RESTful APIs; optimised SQL database queries, reducing average query execution time by ~30%.
- ▶ Participated in Agile sprint cycles, code reviews, and technical documentation, improving team velocity and knowledge-sharing.
- ▶ Collaborated cross-functionally with senior developers to troubleshoot and resolve critical backend issues, reducing bug-resolution time.

IT & Cybersecurity Intern | Ghana Civil Aviation Authority

Nov 2025 – Dec 2025

Accra, Ghana

- ▶ Supported daily maintenance of enterprise IT infrastructure, ensuring >99% system uptime across mission-critical aviation systems.
- ▶ Assisted in implementing security protocols and cybersecurity monitoring frameworks to safeguard sensitive national data.
- ▶ Conducted vulnerability assessments and contributed to risk analysis reports, identifying and flagging high-priority security gaps.
- ▶ Produced detailed IT documentation and technical reports to support audit readiness and knowledge transfer.

System Administrator / IT Technician Intern | TotalEnergies

Mar 2024 – Jun 2024

Marketing Ghana PLC

Accra, Ghana

- ▶ Managed and maintained IT systems for a multinational energy corporation, troubleshooting hardware/software issues and ensuring network security across the office.
- ▶ Installed and configured software environments, delivering technical support to 50+ end-users and optimising system performance.
- ▶ Applied cybersecurity best practices and contributed to backend development tasks that enhanced core IT infrastructure.
- ▶ Collaborated with cross-functional teams to design and implement innovative technology solutions, improving operational efficiency.

Software Developer / Research Analyst Intern | Zormor

Nov 2023 – Mar 2024

Accra, Ghana

- ▶ Developed and maintained software applications, focusing on system performance improvements and security hardening.
- ▶ Conducted data-driven research on emerging technologies and industry trends to inform product roadmap decisions.
- ▶ Analysed datasets to support software development cycles, presenting findings to engineering leads.
- ▶ Collaborated in an agile team to solve technical challenges and deliver enhanced software functionality on schedule.

EDUCATION

BSc Mathematical Sciences (Computer Science with Statistics)

2022 – 2026 (Expected)

University of Ghana, Accra

WASSCE General Science

2019 – 2022

Presbyterian Boys' Secondary School (PRESEC-LEGON), Accra

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- ▶ **Applied Data Science Certificate** — WorldQuant University (In Progress) · Statistical Modelling, Machine Learning, Python for Data Science
- ▶ **Cybersecurity Fundamentals** — IBM SkillsBuild · Cyber threat groups, attack types, social engineering, cryptography, security strategies, and organisational incident response.
- ▶ **ALX Professional Foundations Certificate** — ALX Africa (Sep 2025) · Professional Development Skills for the Digital Age; leadership, communication & problem-solving.

KEY PROJECTS

- ▶ **Pothole Detection & Road Condition Heatmap System (Final Year Project):** Computer vision ML model using OpenCV and deep learning to detect potholes from dashcam footage in real time. Integrates sensor and radar data to generate geospatial heatmaps of poor road conditions, enabling relevant government ministries and road authorities to prioritise and expedite road maintenance across Ghana.

- ▶ **Exoplanet Classification ML Model (NASA Space Apps — Top 2, Ghana):** Machine learning pipeline using NASA's Kepler dataset to identify and classify exoplanets; applied feature engineering, model selection, and performance tuning — ranked Top 2 nationally at the 2025 NASA International Space Apps Challenge.
- ▶ **Expert System for Illegal Mining (Galamsey) in Ghana:** Generative AI logic system built with Python Flask and Prolog to assess environmental pollution risks from illegal mining; integrates rule-based reasoning with a web API for real-time risk evaluation.
- ▶ **Product Z — AI Business Assistant for African SMEs:** Ongoing: AI-powered smart business assistant designed to support African small and medium enterprises with data-driven decision-making, inventory, and customer engagement.
- ▶ **Real-Time Chatbot Application:** Full-stack messaging service with a Node.js backend (WebSockets) and a React frontend; supports live multi-user chat with persistent message history.
- ▶ **NHIS Portal UX Redesign:** UX/UI redesign of the National Health Insurance Scheme portal using React and Next.js, improving accessibility and mobile responsiveness.

ACHIEVEMENTS & COMPETITIONS

- ▶ **NASA International Space Apps Challenge 2025 — Top 2 Team, Ghana:** Led ML development using NASA's Kepler exoplanet dataset; ranked Top 2 nationally and awarded the Galactic Problem Solver distinction by NASA (Oct 4–5, 2025) for outstanding participation and efforts addressing challenges on Earth and in space. Selected to represent Ghana internationally.
- ▶ **UN Quality of Life Hackathon 2025 — Participant, Accra:** Participated in the United Nations-organised hackathon focused on leveraging technology to mitigate daily challenges faced by Ghanaians and improve quality of life through innovative digital solutions.

VOLUNTEER & COMMUNITY INVOLVEMENT

- ▶ **UN Quality of Life Hackathon 2025** — Applied technical skills to real-world civic challenges, collaborating with multidisciplinary teams to prototype technology-driven solutions for Ghanaian communities.
- ▶ **Open-Source & Tech Community** — Actively contributes to collaborative software projects via GitHub; engages with developer communities to stay current with industry trends and best practices.

PROFESSIONAL PROFILE & CAREER INTERESTS

Seeking graduate or junior roles in Backend Engineering, Data Science & Machine Learning, Computer Vision, or Cybersecurity — open to any industry where these skills create impact, including fintech, health-tech, govtech, telecoms, and organisations driving digital transformation across Africa and beyond. Available for hybrid or remote opportunities from June 2026 following degree completion.

LANGUAGES

English (Fluent) **Twɔ** (Fluent) **Fante** (Fluent)

REFERENCES

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